STANet: A Hybrid Spectral and Texture Attention Pyramid Network for Spectral Super-Resolution of Remote Sensing Images

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***Abstract*— Spectral super-resolution (SSR) aims to improve the spectral resolution of images from multispectral imagery or even red, green, blue (RGB) images. However, the majority of existing SSR methods do not fully exploit the spatial and texture features in RGB images, which would lead to the image unreal and distort of the high-frequency details in the reconstructed SSR images. In this study, a hybrid spectral and texture attention pyramid network (STANet) is proposed to reconstruct hyperspectral images (HSIs) with RGB bands of remote sensing images as input. More specifically, a learnable texture feature extraction module is proposed, aiming to make full use of the texture features in the RGB images, which are important in the subsequent spectral reconstruction. Furthermore, to better reconstruct the correlations between various spectral channels, a spatial-spectral-constrained cross-attention module is introduced. Finally, a novel spectral-texture fusion method is proposed, which successfully alleviates the problem of insufficient deep interaction among multiple deep features. On three remote sensing datasets, STANet demonstrates state-of-the-art performance, with its peak signal-to-noise ratio (PSNR) exceeding the suboptimal methods by 0.7266, 0.6724, and 0.6 dB, respectively. The results of the land-cover classification experiment using the reconstructed HSI further demonstrated the performance of the STANet algorithm.**

***Index Terms*— Spectral attention, spectral super-resolution (SSR), spectral-texture fusion, texture attention, texture feature extraction.**

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I. Introduction

HYPERSPECTRAL remote sensing utilizes numerous narrow spectral bands to capture the continuous spectral response of objects, facilitating the detection of subtle differences in various targets. Therefore, hyperspectral remote sensing has found extensive applications in classification [1], [2], target detection [3], and environmental monitoring [4]. However, due to its narrow-band imaging characteristics, hyperspectral remote sensing often exhibits lower spatial resolution [5], [6]. Multispectral imagery always has a higher spatial resolution, which provides rich shape and texture information but limited spectral information. Reconstructing high-resolution hyperspectral data based on limited bands of red, green, blue (RGB) images while maintaining the same spatial resolution has always been a challenging task. Fortunately, spectral super-resolution (SSR) techniques can be used to generate corresponding high-resolution hyperspectral images (HSIs) from RGB images.

SSR could significantly enhance the spectral resolution without compromising spatial resolution [7]. This makes it widely applicable in fields such as remote sensing [8], [9], [10], [11]. Based on whether reference images are used, SSR methods can be categorized into fusion-based SSR and single-image SSR [12], [13], [14]. The fusion SSR method leverages two or more images to obtain HSI. However, this fusion process necessitates that the images be acquired in the same scene, and precise registration is required. Single-image SSR refers to reconstructing images using only a single low-spectral resolution (LR) image. It does not rely on multiple reference image data, making it an efficient solution for achieving image enhancement [15].

However, SSR is a typical ill-posed problem [16]. Various solutions have been proposed to tackle such challenges, and they can be categorized into traditional methods and deep learning-based approaches. Traditional methods mainly depend on priors or assumptions to constrain the spectral reconstruction problem [17]. For instance, Ayala et al. [18] and Zhang and Xu [19] improved principal component analysis (PCA) for the reconstruction of spectral reflectance. They partitioned the spectral space and calculated the principal components for each subspace, which significantly improved

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the accuracy of spectral reflectance estimation. Eslahi et al. [20] implemented spectral recovery using weighted canonical correlation regression. Nguyen et al. [21] utilized a radial basis function network and employed RGB white-balancing to normalize the scene's illumination, thereby recovering the reflectance properties of the scene. In addition to that, sparse learning techniques are also applied in SSR. For example, Arad and Ben-Shahar [22] accomplished fast and accurate reconstruction of HSI by constructing hyperspectral priors and sparse dictionaries. Furthermore, some researchers introduced matrix factorization [23], [24] or spatial constraints (SCs) [25] on this basis to further improve the reconstruction performance. However, such methods exhibit limited performance in feature extraction and poor generalization capabilities, leading to relatively lower accuracy in SSR tasks [16]. Furthermore, given the substantial disparities between the spectral bands of LR and HSI, it becomes a formidable task to accurately represent the spectral characteristics of real objects with limited prior knowledge or assumptions [26], [27].

In recent years, owing to the advantages of deep learning in handling nonlinear mappings, it has been extensively employed in the field of SSR. Among these approaches, those based on convolutional neural networks (CNNs) are the most prevalent and have demonstrated substantial performance enhancements compared with traditional methods [28]. Various architectures and strategies have been proposed to enhance CNN models, such as residual structures [29], [30], [31], multiscale structures [32], [33], [34], and 3-D convolution [35], [36]. Although CNN architectures exhibit commendable performance, a deeper network is often necessitated for handling intricate tasks to further improve performance [37], [38], [39]. However, this may lead to increased computational requirements and the loss of some high-frequency feature information [16]. Additionally, due to the inherent local receptive fields of convolutional networks, there is still room for improvement in extracting global features.

In order to mitigate this constraint, attention mechanisms are adopted in the SSR field. It can enable the model to prioritize the most informative parts. For instance, Zhao et al. [38] proposed a four-level hierarchical regression network (HRNet), which utilized residual blocks to construct an attention mechanism that enlarges the perceptual field. Li et al. [37] presented a novel adaptive weighted attention network (AWAN), redistributing channel feature responses by integrating channel correlation for more accurate reconstruction. The advent of the vision transformer (VIT) further enhances feature extraction and reconstruction performance [40]. Cai et al. [41] proposed a mask-guided spectral-wise transformer (MST) network. They employed masks to guide the restoration of HSI, which enhanced the capability to capture spectral similarity and long-term dependencies. While a majority of attention-based SSR models put forth various solutions for SSR tasks, they have not particularly concentrated on the spatial and texture details of the original images. Neglecting texture information in SSR model design may result in insufficiently capturing subtle but crucial variations between adjacent pixels, potentially leading to the omission of critical details during the SSR process. Furthermore, the utilization of deeper feature extraction architectures aids in capturing essential feature information. However, it also leads to the loss of certain high-frequency fine details and requires greater computational resources. This, in turn, unavoidably impacts the enhancement of model performance.

To address the aforementioned issues, a spectral-texture attention-based super-resolution network is proposed. First, to facilitate a more exhaustive extraction and reconstruction of spectral features, the spatially guided spectral attention mechanism is improved to aggregate spectral features. A cross-attention module based on spatial-spectral constraints is incorporated, with the objective of augmenting spectral reconstruction while alleviating computational load to some extent. Subsequently, to fully harness the texture and spatial features present in the original RGB images, a trainable texture feature extraction module along with a multiscale network structure is introduced. This facilitates a comprehensive exploration of spatial and structural information within original images. Lastly, considering that the features from diverse types and scales fail to fully explore and exploit the potential correlations between features, the texture attention structure and the deep feature fusion block are designed. This allows for more effective feature fusion through the introduction of deep interactions.

The main contributions of this article are as follows.

1) A hybrid spectral and texture attention pyramid network (STANet) is proposed for HSI reconstruction from RGB images, aiming to integrate spatial and texture information as constraints during the spectral enhancement process. By precisely perceiving and leveraging high-frequency details and local spatial similarity, the method enhances its capability to restore complex textures and local features in the image.
2) A spatial-guided spectral attention module is presented to enhance spectral information while preserving spatial features, achieving spectral consistency within the spatial context in reconstructed images. Additionally, a texture-guided adaptive attention structure is introduced to inherit high-frequency features from the original RGB image, alleviating potential detail distortion in the reconstructed image.
3) STANet demonstrates superior spectral fidelity and excellent spatial detail and has exhibited state-of-the-art performance across three datasets. A land use classification application further validates the practicality of the HSI generated by our proposed methods.

The remaining sections of this article are organized as follows. Section II discusses our proposed STANet network structure in detail. In Section III, our experimental environment and the experimental results are introduced. Finally, the conclusion is given in Section IV.

II. Methods

A. Overview of STANet

Fig. 1 illustrates the overall architecture of the STANet, which primarily consists of ConvNeXt blocks, spatially guided spectral attention module, texture attention module (TAM),

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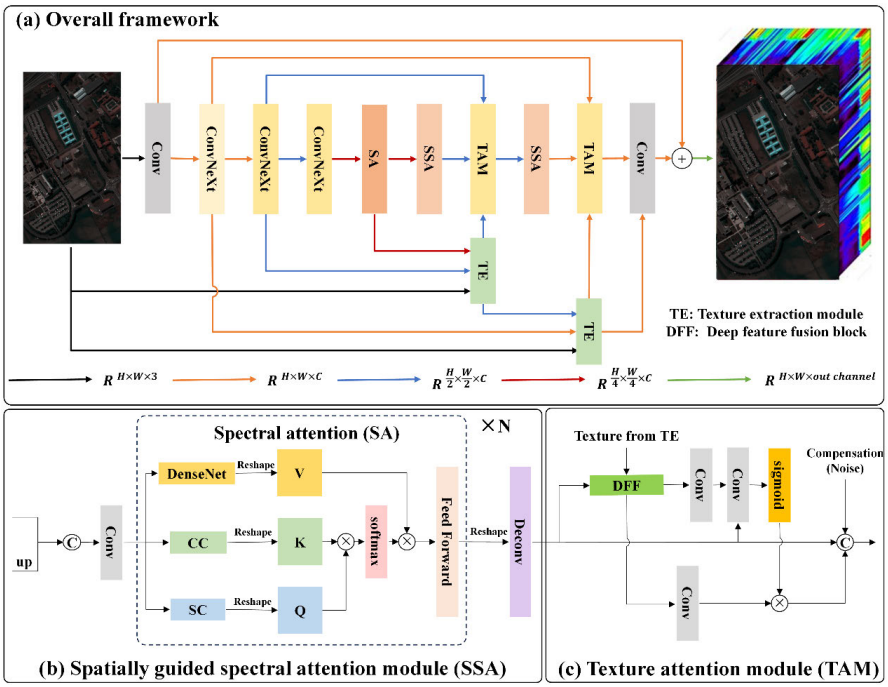


Fig. 1. (a) Overall framework, (b) spectral attention, and (c) TAM. (*Note:* the downsampling function is not utilized in the initial ConvNeXt block, which is represented in light orange color.)

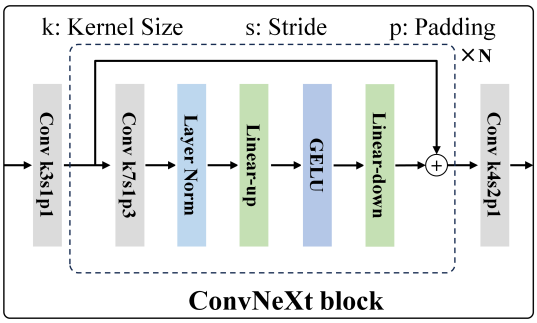


Fig. 2. ConvNeXt block.

and texture extraction module. First, STANet uses ConvNeXt blocks for feature extraction and downsampling to obtain deep features at three different scales ($R^{H\times W\times C}$(Height×Width×Channel), $R^{(H/2)\times(W/2)\times C}$, $R^{(H/4)\times(W/4)\times C}$). The structure of the ConvNeXt block is shown in Fig. 2 and can be represented as follows:

$$\mathbf{D_1} = f_{\text{ConvX}}(\text{Conv2}d(\mathbf{I}_{\text{LR}})) \tag{1}$$

$$\mathbf{D_2} = f_{\text{ConvX}}(\mathbf{D_1}) \tag{2}$$

$$\mathbf{D_3} = f_{\text{ConvX}}(\mathbf{D_2}) \tag{3}$$

where f_{ConvX} represents the ConvNeXt block. Conv2d denotes the convolution operation, $\mathbf{I}_{\text{LR}} \in R^{H\times W\times 3}$ represents the input LR RGB images, and $\mathbf{D}_i$ represents deep features at different scales obtained using the ConvNeXt block. $\mathbf{D_1}$ represents scale-invariant deep features; $\mathbf{D_2}$ and $\mathbf{D_3}$ represent deep features obtained after a 2× and 4× downsampling ($\mathbf{D_1} \in R^{H\times W\times C}$, $\mathbf{D_2} \in R^{(H/2)\times(W/2)\times C}$, $\mathbf{D_3} \in R^{(H/4)\times(W/4)\times C}$).

Subsequently, the features at different scales are processed sequentially, starting with spectral aggregation applied to the deep features $\mathbf{D_3}$. Only one spectral attention operation is used here for preliminary extraction, represented as the SA operation in Fig. 1(a)

$$\mathbf{S_1} = f_{\text{SA}}(\mathbf{D_3}) \tag{4}$$

where f_{SA} represents the spectral attention operation SA, and $\mathbf{S_1} \in R^{(H/4)\times(W/4)\times C}$ corresponds to the extracted spectral features.

Afterwards, the obtained spectral aggregation features $\mathbf{S_1}$ are subjected to texture extraction and spectral cross-attention operations separately. This can be represented as follows:

$$\mathbf{T_1} = f_{\text{TE}}(\text{up}(\mathbf{S_1}), \mathbf{D_2}, \text{down}(\mathbf{I_{LR}})) \tag{5}$$

$$\mathbf{S_2} = f_{\text{SSA}}(\text{up}(\mathbf{S_1}), \mathbf{D_2})) \tag{6}$$

where f_{TE} represents the texture extraction module, f_{SAN} represents spatially guided spectral attention module, $\mathbf{T_1} \in R^{(H/2)\times(W/2)\times C}$ corresponds to the extracted texture features, and $\mathbf{S_2} \in R^{(H/2)\times(W/2)\times C}$ represents the extracted spectral features. up and down represent 2× upsampling and 2× downsampling operations, respectively.

Then, the obtained texture features $\mathbf{T_1}$ and spectral features $\mathbf{S_2}$ are fused using a TAM to obtain fusion features $\mathbf{A_1} \in R^{(H/2)\times(W/2)\times C}$ that incorporate both well-integrated texture and spectral information. This can be represented as follows:

$$\mathbf{A_1} = f_{\text{TA}}(\mathbf{S_2}, \mathbf{T_1}) \tag{7}$$

where f_{TA} represents the TAM. $\mathbf{A_1}$ represents the spectral-texture fusion features.

Subsequently, the fusion features $\mathbf{A_1}$ undergo the same operation as those applied to $\mathbf{S_1}$. This can be represented as

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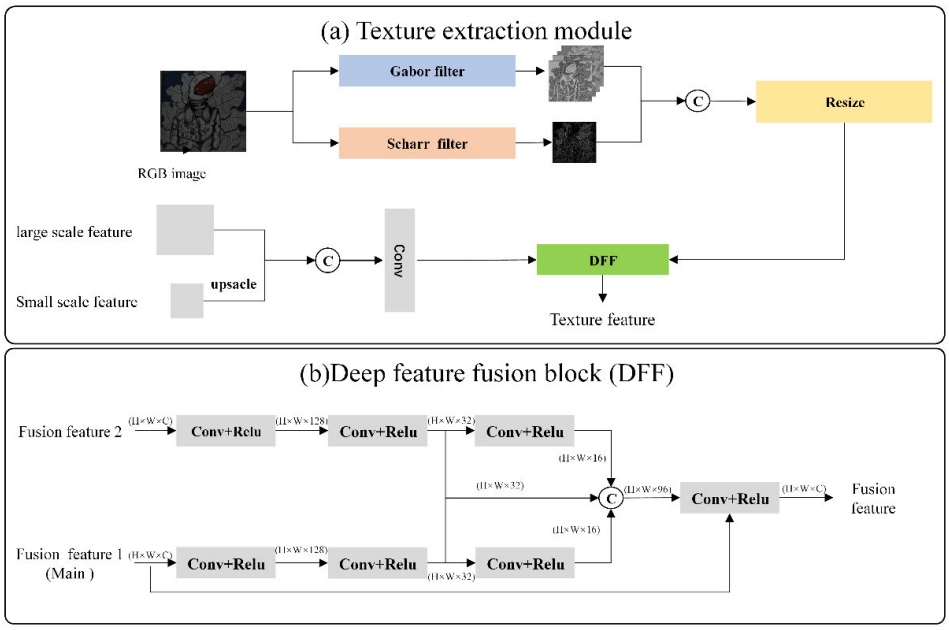


Fig. 3. (a) Texture extraction module and (b) deep feature fusion block.

follows:

$$\mathbf{T_2} = f_{\mathrm{TE}}(\mathrm{up}(\mathbf{T_1}), \mathbf{D_1}, \mathbf{I_{LR}}) \tag{8}$$

$$\mathbf{S_3} = f_{\mathrm{SSA}}(\mathrm{up}(\mathbf{A_1}), \mathbf{D_1})) \tag{9}$$

$$\mathbf{A_2} = f_{\mathrm{TA}}(\mathbf{S_3}, \mathbf{T_2}). \tag{10}$$

Finally, texture features $\mathbf{T_2} \in R^{H\times W\times C}$ and fusion features $\mathbf{A_2} \in R^{H\times W\times C}$ are concatenated to obtain the ultimate reconstruction result $\mathbf{I}_{\mathrm{SSR}}$

$$\mathbf{I}_{\mathrm{SSR}} = \mathrm{Conv2}d(\mathrm{concat}(\mathbf{T_2}, \mathbf{A_2})) + \mathrm{Conv2}d(\mathbf{I}_{\mathrm{LR}}) \tag{11}$$

in which concat represents the concatenation operation.

B. Proposed STANet

1) ConvNeXt Block: The pyramid network capitalizes on multiscale information for feature extraction and processing. This enables the network to have an advantage in handling complex reconstruction tasks. Hence, our model adopts the pyramid network architecture [42], [43]. Initially, we devised a downsampling deep feature extraction structure based on ConvNeXt, known as the ConvNeXt block. The structure is illustrated in Fig. 2 (the initial ConvNeXt block does not include downsampling operations, with the k3s1p1 convolution operation). ConvNeXt is an improved ResNet architecture that draws inspiration from the Swin transformer architecture. It enhances feature extraction capability while maintaining a relatively low parameter computation load [44].

2) Texture Extraction Module: In the SSR tasks, ignoring image texture features could potentially impose limitations on the precision of the SSR model [16]. Therefore, we introduce a learnable texture extraction structure combined with convolutional Gabor filtering and gradient Scharr filtering. The module structure is illustrated in Fig. 3(a).

First, we employ Gabor and gradient Scharr filtering to capture the texture details of LR images. Subsequently, we concatenate these results, which can be expressed in the following:

$$\boldsymbol{t_i} = \mathrm{concat}(\mathrm{Gabor}(\mathbf{I}_{\mathrm{LR}}), \mathrm{Sobel}(\mathbf{I}_{\mathrm{LR}})). \tag{12}$$

$\boldsymbol{t_i}$ represents the extracted texture information, with Gabor and Sobel denoting Gabor filter and gradient Scharr filter, respectively. The calculation formulas and principles of Gabor filter and gradient Scharr filter are as follows.

a) Gabor filter: Gabor filters can utilize an improved trainable structure to compute texture features at different orientations and scales of the image, thereby better capturing and representing the texture information of the image. The calculation formula for Gabor filters is as follows:

$$\mathrm{Gabor}(x, y; \sigma, \theta, \lambda, \gamma, \psi) = e^{-\frac{x'^2+\gamma^2y'^2}{2\sigma^2}} e^{i\left(\frac{2\pi x'}{\lambda}+\psi\right)} \tag{13}$$

where (x, y) are the original coordinates, and (x', y') are the rotated coordinates. $x' = x\cos\theta + y\sin\theta$, $y' = -x\sin\theta + y\cos\theta$. i stands for the imaginary unit, used to denote the imaginary part, while $e^{i(((2\pi x')/\lambda)+\psi)}$ represents the exponential form of a complex number. σ, θ, λ, γ, and ψ are parameters of the Gabor filter, used to control the characteristics of the filter. The meanings and roles of these parameters are shown in Table I.

In our architecture, the parameters σ, θ, λ, γ, and ψ of this formula are trainable and optimized during the training process. Here, $\mathrm{Gabor}(x, y; \sigma, \theta, \lambda, \gamma, \psi)$ represents the response value of the Gabor filter at position (x, y).

b) Gradient Scharr filter: The gradient texture extraction utilizes Scharr convolution kernels to compute the gradient components of an image in both horizontal (x) and vertical (y) directions. It calculates the gradient magnitude of the image to extract texture details and edge information. Specifically, it first applies Scharr convolution operations separately to the input image $\mathbf{I}_{\mathrm{LR}}$ in the x and y-directions, resulting in the corresponding x-direction gradient component g_{xx}, y-direction

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TABLE I
MEANINGS AND ROLES OF THE PARAMETERS

Symbol	Physical Meaning	Purpose
σ	Bandwidth	Variance of the Gaussian filter, controls the size of texture
θ	Orientation	Determines the directionality of the texture
λ	Wavelength	Controls the frequency of the texture
γ	Spatial Aspect Ratio	Adjusts the filter's spatial shape
ψ	Phase Offset	Modulates the phase offset of the function, controlling filter phase characteristics

gradient component g_{yy}, and the mixed components g_{xy} in both the x and y-directions

$$g_{xx} = \text{conv2}d(\mathbf{I}_{\text{LR}}, \text{kernel}_x) \tag{14}$$

$$g_{yy} = \text{conv2}d(\mathbf{I}_{\text{LR}}, \text{kernel}_y) \tag{15}$$

$$g_{xy} = g_{xx} * g_{yy} \tag{16}$$

in which conv2d represents the convolution operation, and kernel_x and kernel_y represent the Scharr convolution kernels in the horizontal and vertical directions, which can be expressed as

$$\text{kernel}_x = \begin{bmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{bmatrix} \tag{17}$$

$$\text{kernel}_y = \begin{bmatrix} -3 & -10 & -3 \\ 0 & 0 & 0 \\ 3 & 10 & 3 \end{bmatrix}. \tag{18}$$

Afterward, we calculate the gradient magnitude G and gradient direction θ

$$\theta = 0.5 \cdot \text{atan}\left(\frac{2 \cdot g_{xx}}{g_{xx} - g_{yy} + \text{eps}}\right) \tag{19}$$

$$G = (g_{xx} + g_{yy}) + (g_{xx} - g_{yy}) \cdot \cos(2\theta) + 2 \cdot g_{xy} \cdot \sin(2\theta) \tag{20}$$

where atan represents the arctangent function, and eps is a small constant (e.g., 10^{-12}) used for numerical stability.

Finally, the gradient magnitude is obtained, and the enhanced gradient magnitude f is calculated

$$f = \sqrt{0.5 \cdot \max(G, 0)}. \tag{21}$$

$\max(G, 0)$ represents taking the maximum value between G and 0.

While various architectures can provide rich feature information, the majority of deep learning models typically employ traditional feature fusion structures that are confined to rudimentary operations such as addition or concatenation. These simplistic fusion methods exhibit certain performance constraints and do not sufficiently explore and leverage the potential correlations among diverse feature types. Hence, we present a novel deep feature fusion block that facilitates more efficacious feature fusion by introducing depth interactions among features.

Therefore, we use convolutional operations to transform the channel dimensions of fused features, obtaining multidimensional feature. Through the integration of information from different dimensions, we effectively enhance the input features, thereby improving their representational capacity and diversity. The deep feature fusion structure is illustrated in Fig. 3(b).

We fuse deep features $\mathbf{D_i}$ and $\mathbf{D_{i+1}}$ of two scales, while combining the previously extracted texture information $\boldsymbol{t_i}$ to achieve an effective fusion operation, and they can be represented as follows:

$$\mathbf{T_i} = \text{DFF}(\text{Conv2}d(\text{concat}(\text{up}(\mathbf{D_i}), \mathbf{D_{i+1}})), \text{res}(\boldsymbol{t_i})) \tag{22}$$

where DFF represents the deep feature fusion block, res denotes the resizing operation, aimed at adjusting the texture information $\boldsymbol{t_i}$ to the same size as $\mathbf{D_{i+1}}$, and up represents the upsampling operation, and $\mathbf{T_i}$ represents texture features.

3) Spatially Guided Spectral Attention Module: Accurate restoration of spectral information is crucial in the context of SSR. Therefore, efficiently harnessing the advantages of deep learning to reconstruct spectral information becomes particularly important. Currently, multihead attention networks have shown significant promise in spectral reconstruction [41], [45]. However, their substantial computational demands severely impact performance [46]. Moreover, the majority of transformer models do not specifically focus on the reconstruction of spectral information while preserving spatial details.

To address this issue, a spatial-guided cross-attention mechanism based on spectral dimensions is proposed to achieve high-performance spectral reconstruction with lower computational complexity. It reduces computational by basing the attention computation on the channel dimension. The module's structure is illustrated in Fig. 1(b), where small-scale spectral features $\mathbf{S_i}$ are first concatenated with deep features $\mathbf{D_i}$

$$\mathbf{S_i'} = \text{concat}(\text{up}(\mathbf{S_i}), \mathbf{D_i}). \tag{23}$$

Spectral attention and spatial attention modules in convolutional block attention module (CBAM) model [47] are separated and regraded as the SC unit and the spectral constraint (CC) unit, respectively. The results of them are the query $\mathbf{Q_i}$ key $\mathbf{K_i}$. Concurrently, we adopt a dense residual structure [48] (densenet) to thoroughly extract deep feature, resulting in the value $\mathbf{V_i}$. The structures of SC, CC, and dencenet are shown in Fig. 4, and they can be represented by the following formulas:

$$\mathbf{Q_i} = \text{reshape}\big(\text{SC}\big(\text{conv}\big(\mathbf{S_i'}\big)\big)\big) \tag{24}$$

$$\mathbf{K_i} = \text{reshape}\big(\text{CC}\big(\text{conv}\big(\mathbf{S_i'}\big)\big)\big) \tag{25}$$

$$\mathbf{V_i} = \text{reshape}\big(\text{densenet}\big(\text{conv}\big(\mathbf{S_i'}\big)\big)\big). \tag{26}$$

The purpose of reshape is to ensure that the subsequent calculations of multihead attention are performed along the spectral dimension, aiming to reduce computational complexity, results into $\mathbf{Q_i} \in R^{HW \times C}$, $\mathbf{K_i} \in R^{HW \times C}$, $\mathbf{V_i} \in R^{HW \times C}$. Finally, the spectral feature $\mathbf{S_{i+1}}$ is obtained from $\mathbf{Q_i}$, $\mathbf{K_i}$, and $\mathbf{V_i}$ using attention mechanism

$$\mathbf{S_{i+1}} = \text{Deconv}\left(\text{reshape}\left(\left(\text{softmax}\left(\frac{\mathbf{Q_i}\mathbf{K_i^T}}{\sqrt{d_k}}\right)\right)\mathbf{V_i}\right)\right) \tag{27}$$

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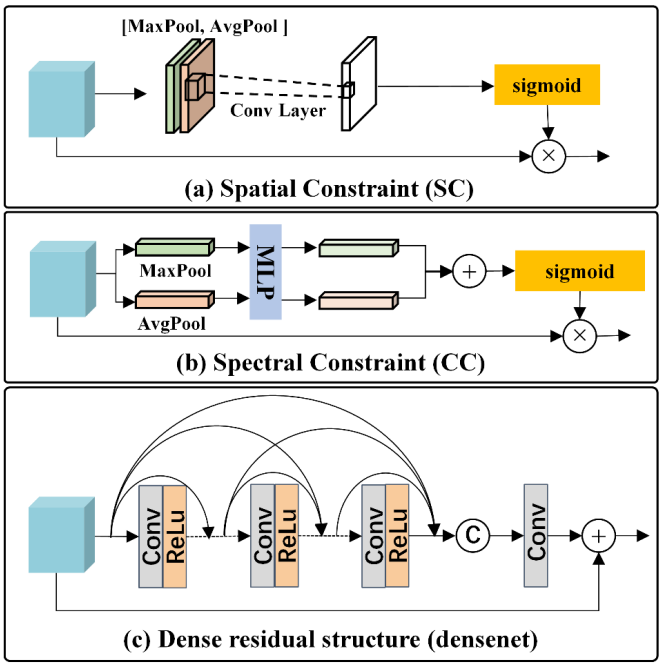


Fig. 4. (a) SC unit, (b) spectral constraint unit (CC), and (c) dense residual structure (densenet).

where softmax refers to the softmax function. d_k represents the dimension of $\mathbf{Q_i}$ and $\mathbf{K_i}$, and its primary function is to scale the attention weights, thereby helping to improve the robustness of the model. Deconv represents deconvolution, thereby achieving 2× upsampling.

4) TAM: The adaptive texture attention mechanism is introduced, aiming to enhancing the spectral information while preserving texture feature of LR images. Additionally, a trainable compensation mechanism is incorporated to realize subtle modification of TAM output. This can enhance both the spatial and spectral details of the reconstructed features. The structure is shown in Fig. 1(c).

First, the spectral features $\mathbf{S_{i+1}}$ are fused with texture features $\mathbf{T_i}$ using deep feature fusion block to obtain fused features $\mathbf{T_i'}$, which can be represented as follows:

$$\mathbf{T_i'} = \mathrm{DFF}(\mathbf{T_i}, \mathbf{S_{i+1}}). \tag{28}$$

Next, by introducing the convolution and attention structure, we have constructed an adaptive attention extraction structure. This structure allows for a flexible capture of crucial information and realizes accurate information extraction. The compensation mechanism is achieved by setting an optimizable weight. Specifically, this process can be described as follows:

$$\mathbf{A_i} = \boldsymbol{Conv}(\mathbf{T_i'}) \odot \boldsymbol{\sigma}\left(\boldsymbol{Conv}\left(\boldsymbol{Conv}(\mathbf{T_i'}) + \mathbf{S_{i+1}}\right)\right) + \mathbf{T_i'} + \mathbf{w} \cdot \boldsymbol{comp} \tag{29}$$

where $\mathbf{A_i}$ represents the features obtained through adaptive texture attention, $\odot$ denotes element-wise multiplication, and $\boldsymbol{\sigma}$ represents the sigmoid function. $\mathbf{w}$ represents a learnable weight of the compensation mechanism, while $\boldsymbol{comp}$ denotes a random noise matrix with the same dimensions as $\mathbf{S_{i+1}}$.

C. Loss Function

Mean relative absolute error (MRAE) is employed as the loss function for HSI restoration. MRAE is a loss function used to assess the performance of prediction models and is widely applied in SSR tasks. It measures the average magnitude of relative absolute errors between the predicted values $\mathbf{I_{SSR}}$ and the ground truths $\mathbf{I_{GT}}$. The MRAE loss function quantifies model performance based on the relative values of absolute errors and can be represented by the following formula:

$$\boldsymbol{loss_{MRAE}} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\left|\mathbf{I}_i^{\mathrm{GT}} - \mathbf{I}_i^{\mathrm{SSR}}\right|}{\mathbf{I}_i^{\mathrm{GT}}}\right). \tag{30}$$

In the involved dataset, some pixels have values of 0 or close to 0. In such cases, using MRAE as the loss function would lead to extremely large or even NA values. Therefore, we have opted to use L1 loss as the loss function, which can be represented as follows:

$$\boldsymbol{loss_{l1}} = \frac{1}{N}\sum_{i=1}^{N}\left(\left|\mathbf{I}_i^{\mathrm{GT}} - \mathbf{I}_i^{\mathrm{SSR}}\right|\right). \tag{31}$$

The remote sensing image datasets in this study utilize the L1 loss function, whereas the natural image datasets predominantly employ the MRAE loss function (please refer to the supplementary material).

III. Experimental Results and Analysis

A. Experimental Setup

1) Dataset: This study investigates the performance of the STANet on three publicly available remote sensing datasets, including the Pavia University scene (PaviaU) dataset, the Chikusei dataset [49], and the Yellow River Estuary (YRE) dataset. In these datasets, RGB images were obtained based on the spectral response functions of Sentinel-2 [50]. These datasets encompass a wide range of remote sensing scenarios. We also conducted relevant experiments on natural images, and the details can be found in the supplementary material.

The PaviaU dataset is extensively used for remote sensing HSI classification and segmentation, collected from the University of Pavia in Italy. With a spatial resolution of 1.3 m, the dataset comprises 103 bands covering the wavelength range of 0.43–0.86 μm and has a data size of 610 × 340 pixels. It encompasses nine main land-cover types. We create corresponding RGB images using the spectral response functions of the Sentinel-2 satellite. For training, we selected the upper segment of the image, sized at 300 × 340 pixels, which was then subdivided into 32 × 32 blocks. The remaining portion was used for validation.

The Chikusei dataset consists of HSI obtained from the Chikusei region in Ibarakiken, Japan. The dataset exhibits a spatial resolution of 2.5 m and a size of 2517 × 2335 pixels. It includes a total of 128 spectral bands, spanning the spectral range from 363 to 1018 nm. To focus on the central area and exclude irregular portions at the edges, we perform dataset cropping, resulting in a central portion with dimensions of 2304 × 2176 pixels. We create corresponding RGB images

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using the spectral response functions of the Sentinel-2 satellite. Finally, we crop these images into small patches of size 512 × 128 pixels, generating a total of 72 images. Among these, 62 are used for training, and the remaining ten are used for testing.

The YRE dataset was captured by the GaoFen-5 B (GF-5B) satellite on November 1, 2018, in the YRE area of Dongying City, Shandong Province, China. The hyperspectral data have a spatial resolution of 30 m, covering 318 bands in the wavelength range of 0.4–2.5 μm, with a data size of 1064 × 1548 pixels. Twenty bad bands were eliminated, resulting in 298 band data. The corresponding RGB images are created using the spectral response functions of the Sentinel-2 satellite. For training, we selected the upper segment of the image, with size of 700 × 1548 pixels, which was then subdivided into 32 × 32 blocks. The remaining part was used for validation.

2) Model Evaluation: To more accurately assess the model's performance, both quantitative and qualitative indicators are employed. As for the qualitative assessment, our evaluation primarily involves comparing local zoomed-in RGB visual images, image universal quality index (UQI) maps [51], mean square error (MSE) maps, and spectral curve plots. Specifically, MSE maps are generated by calculating the MSE for each pixel between the reconstructed images and the ground truth images. This helps quantify pixel-level differences between the reconstructed and ground truth images (smaller MSE values indicate better performance). UQI maps are generated by comparing the quality differences between the reconstructed images and the original images. This helps understand the overall similarity between the reconstructed and original images (greater value index indicates better performance). Local zoomed-in images are created by selecting specific spectral bands to reconstruct RGB images, demonstrating differences in how various models handle image details. This aids in observing the performance of different models in detail reconstruction. Spectral curve plots involve selecting specific points within the images and comparing the spectral curves of the reconstructed images generated by various methods. This helps us understand performance differences among different models in different spectral bands (greater correlation indicates better performance). Using these validation methods enables a more comprehensive evaluation of our model's performance and results. Additionally, remote sensing land-cover classification experiments are conducted on the reconstructed hyperspectral data to demonstrate the effectiveness of our experimental results. In terms of quantitative approaches, the following methods are utilized.

1) MSE

$$\mathrm{MSE}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}}) = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{I}_i^{\boldsymbol{GT}} - \mathbf{I}_i^{\boldsymbol{SSR}}\right)^{\mathbf{2}}. \tag{32}$$

2) Peak signal-to-noise ratio (PSNR)

$$\mathrm{PSNR}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}}) = 10 \cdot \boldsymbol{log}_{10}\frac{\mathrm{Max}^2}{\mathrm{MSE}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}})}. \tag{33}$$

TABLE II
COMPARISON MODEL USED IN THIS ARTICLE

Model	Competition	Track	Ranking
EDSR [28]	NTIRE 2017	Super-resolution	First place
HSCNN+ [31]	NTIRE 2018	Spectral Reconstruction Challenge	First place
HRNET [38]	NTIRE 2020	Spectral Reconstruction Challenge (Real World track) and (Clean track),	First place and third place
AWAN [37]	NTIRE 2020	Spectral Reconstruction Challenge (Real World track) and (Clean track),	Third place and First place
HINET [34]	NTIRE 2021	Image Deblurring JPEG Artifacts track	First place
MST++ [41]	NTIRE 2022	Spectral Reconstruction Challenge	First place

3) Structure similarity index measure (SSIM)

$$\mathrm{SSIM}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}}) = \frac{(2\mu_{\mathbf{I}_{\mathrm{SSR}}}\mu_{\mathbf{I}_{\mathrm{GT}}} + C_1)(2\sigma_{\mathbf{I}_{\mathrm{SSR}}\mathbf{I}_{\mathrm{GT}}} + C_2)}{(\mu^2_{\mathbf{I}_{\mathrm{SSR}}} + \mu^2_{\mathbf{I}_{\mathrm{GT}}} + C_1)(\sigma^2_{\mathbf{I}_{\mathrm{SSR}}} + \sigma^2_{\mathbf{I}_{\mathrm{GT}}} + C_2)}. \tag{34}$$

4) Spectral angle measure (SAM)

$$\mathrm{SAM}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}}) = \arccos\frac{\mathbf{I}_{\mathrm{SSR}} \cdot E\mathbf{I}_{\mathrm{GT}}}{\|\mathbf{I}_{\mathrm{SSR}}\| \times \|\mathbf{I}_{\mathrm{GT}}\|}. \tag{35}$$

5) Erreur relative globale adimensionnelle de synthèse (ERGAS)

$$\mathrm{ERGAS}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}}) = 100p \cdot \sqrt{\frac{1}{N}\sum_{i=1}^{N}\frac{\|\mathbf{I}_{\mathrm{GT}} - \mathbf{I}_{\mathrm{SSR}}\|_2^2}{\left(\frac{1}{D}1_D^t\mathbf{I}_{\mathrm{GT}}\right)^2}}. \tag{36}$$

6) UQI

$$\mathrm{UQI}(\mathbf{I}_{\mathrm{SSR}}, \mathbf{I}_{\mathrm{GT}}) = \frac{\mathbf{4} \cdot \sigma_{\mathbf{I}_{\mathrm{SSR}}\mathbf{I}_{\mathrm{GT}}} \cdot \mu_{\mathbf{I}_{\mathrm{SSR}}} \cdot \mu_{\mathbf{I}_{\mathrm{GT}}}}{(\mu^2_{\mathbf{I}_{\mathrm{SSR}}} + \mu^2_{\mathbf{I}_{\mathrm{GT}}})(\sigma^2_{\mathbf{I}_{\mathrm{SSR}}} + \sigma^2_{\mathbf{I}_{\mathrm{GT}}})}. \tag{37}$$

STANet is compared with several state-of-the-art methods, including EDSR [28], HSCNN+ [31], HRNET [38], AWAN [37], HINET [34], and MST++ [41]. These methods are introduced in the Introduction part. These methods are the excellent solutions in the new trends in image restoration and enhancement (NTIRE) competition, with specific details shown in Table II. Besides, we have also incorporated the two latest super-resolution algorithms, SSRnet [52] and LTRN [53].

3) Implementation Details: For optimization, Adam optimizer [54] is used to train the model, where $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$. The learning rate is set to 4×10^{-4}. The total training epoch is 300. A computer with NVIDIA GTX3080TI GPU and AMD Ryzen 9 5950X CPU is used. The model is tested on pytorch using python3 under windows 10, cuda 11.7, and cudnn 8.4.0 systems. The parameters of other comparison methods are set to their default.

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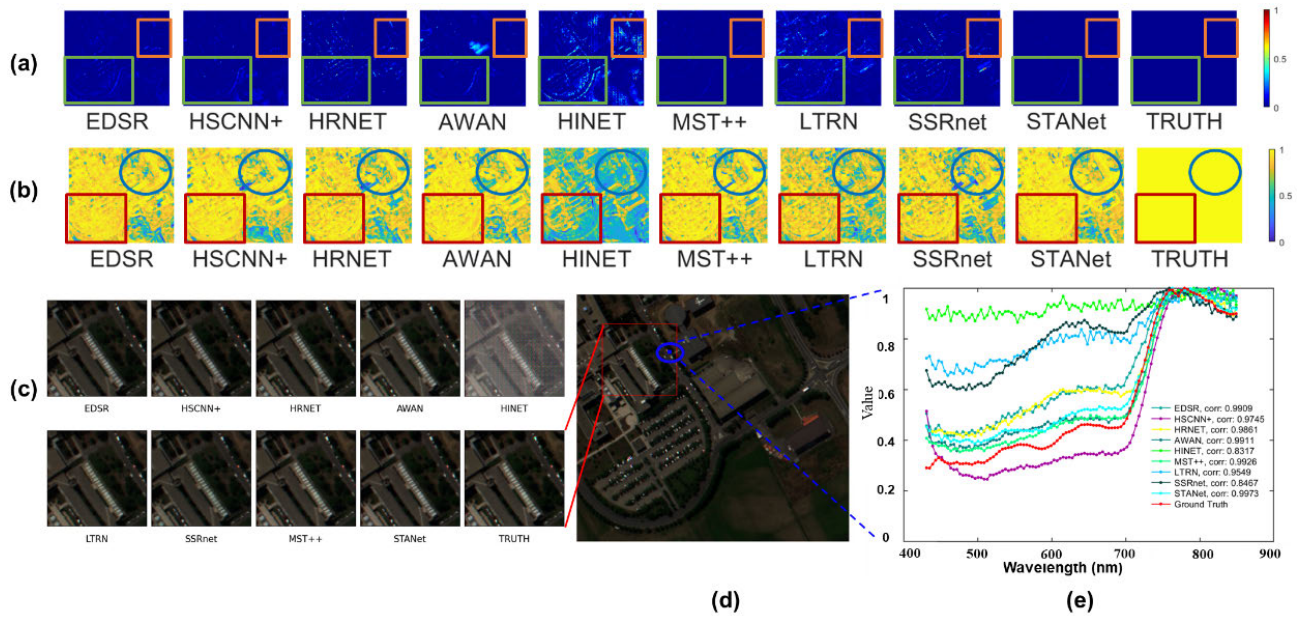


Fig. 5. Qualitative results of the PaviaU dataset. (a) MSE maps and (b) UQI maps for the spectral band (the 90th band is used here for example). The local zoom-in images (c) created by combining bands of 14, 31, and 57 as the R/G/B channels from the constructed HSI. (d) Displays the original RGB image. The spectral curve (e) of the blue region in recovered and original HSI, as well as the correlation coefficient with the reference spectral curve.

TABLE III
QUANTITATIVE RESULTS OF PAVIAU DATASET

	PNSR↑	SSIM↑	ERGAS↓	SAM↓
EDSR	32.7432	0.9776	1.7178	4.3746
HSCNN+	30.5469	0.962	2.2301	5.1119
AWAN	31.3821	0.9681	2.0298	5.0409
HRNET	30.4192	0.9576	2.4279	5.5167
HINET	19.9449	0.6968	7.9575	23.5549
SSRnet	29.9535	0.9543	2.4105	6.1487
LTRN	31.4167	0.9441	2.4836	5.2181
MST++	32.8081	0.9775	1.706	4.3128
STANet	**33.5347**	**0.981**	**1.5633**	**3.885**

TABLE IV
QUANTITATIVE RESULTS OF CHIKUSEI DATASET

	PNSR↑	SSIM↑	ERGAS↓	SAM↓
EDSR	26.6997	0.9161	4.3502	10.8074
HSCNN+	21.9452	0.8601	6.5453	9.0452
AWAN	27.5749	0.9356	3.3395	7.1482
HRNET	27.7112	0.927	3.9081	10.855
HINET	29.5408	0.9648	2.5409	7.166
SSRnet	28.4213	0.9542	3.0644	8.8492
LTRN	25.0733	0.9030	5.1289	6.9354
MST++	30.2784	0.9681	2.4065	6.2871
STANet	**30.9526**	**0.9744**	**2.1557**	**6.0579**

TABLE V
QUANTITATIVE RESULTS OF YRE DATASET

	PNSR↑	SSIM↑	ERGAS↓	SAM↓
EDSR	29.9817	0.7426	3.3633	12.2856
HSCNN+	28.0735	0.7284	4.033	13.6412
AWAN	28.0617	0.7837	3.1503	12.441
HRNET	29.4799	0.6989	3.7375	13.6828
HINET	24.6993	0.5842	4.6369	15.3295
SSRnet	28.5437	0.6766	3.3869	13.5664
LTRN	30.2616	0.7182	3.1478	11.0423
MST++	28.3075	0.7583	3.2947	11.1522
STANet	**30.5817**	**0.8003**	**2.9975**	**10.54**

B. Experimental Results

1) Implementation Details: Tables III–V present the quantitative results of SSR on the PaviaU dataset, the Chikusei dataset, and the YRE dataset, respectively. Bold font indicates the optimal results, while underlined values indicate the suboptimal results. Overall, our method achieved the optimal accuracy on all three datasets, demonstrating the superiority of STANet.

On the PaviaU dataset, STANet surpasses the suboptimal method, MST++, with a margin of 0.7266 dB and 0.0035 in terms of the PSNR and SSIM metrics, respectively. This indicates that STANet has advantage in maintaining image quality and similarity. Additionally, the ERGAS and SAM of STANet decrease 0.1427 and 0.4278, respectively, implying its ability to better preserve spectral accuracy and similarity.

Similar phenomena were witnessed on the Chikusei dataset and YRE dataset. The proposed STANet algorithm outperformed the comparative algorithms in all evaluation metrics. This further validates the superiority of the algorithm proposed in our study.

2) Qualitative Comparison: Four different indicators are employed to validate the effectiveness of our model, including MSE maps, UQI maps, local zoomed-in images, and spectral curve plots. Due to computational constraints, the PaviaU dataset and TRE dataset display all regions of the validation images, while the Chikusei dataset present validation images of sizes 128 × 512.

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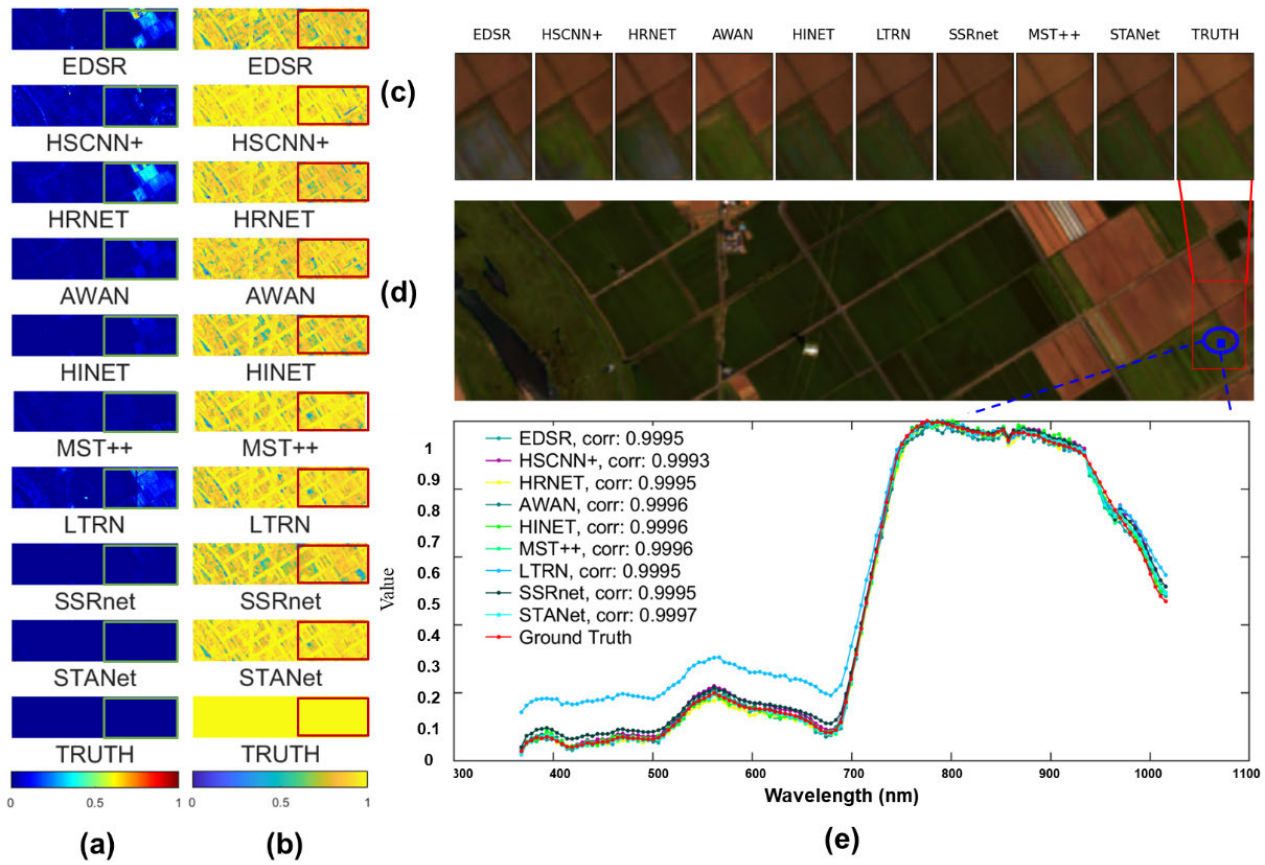


Fig. 6. Qualitative results of the Chikusei dataset. (a) MSE maps and (b) UQI maps for the spectral band (the 40th band is used here for example). (c) Local zoom-in images created by combining bands of 13, 35, and 65 as the R/G/B channels from the constructed HSI. (d) Displays the original RGB image. The spectral curve (e) of the blue region in recovered and original HSI, as well as the correlation coefficient with the reference spectral curve.

Fig. 5 displays comparative results from the PaviaU dataset. It is evident that STANet outperforms the comparison algorithms in all four-evaluation metrics. Specifically, the MSE maps [Fig. 5(a)] reveal that STANet exhibits superior reconstruction performance in the self-blocking bricks (inside green box) and the road (orange box) compared to other methods. The UQI maps [Fig. 5(b)] demonstrate that STANet excels in the reconstruction quality of the bare soil (inside blue ellipse), self-blocking bricks (inside red box), and background compared to other methods. In the local zoomed-in images [Fig. 5(c)], we selected the vegetation area as the comparison area, as it contains rich detail information. It can be observed that STANet exhibits great performance. In the spectral curve plots, we selected an area of 4 × 4 pixel for comparison [Fig. 5(e)]. The results show that STANet's reconstruction has the highest correlation with the original HSI. This further validates the excellent performance of STANet in spectral reconstruction.

Fig. 6 presents comparative results of the Chikusei dataset. STANet outperforms other methods in reconstructing the right part of the image [Fig. 6(a)], particularly the bare soil area (inside green box). It is worth noting that in the UQI maps [Fig. 6(b)], STANet performs better in the reconstruction of different adjacent plots. In local zoomed-in images [Fig. 6(c)], it can be observed that STANet does not exhibit color shifts in the reconstructed agricultural plots. The colors of the plot reconstructions appear reasonable, and the texture of the features within the plots is also well reconstructed. In spectral curve plots [Fig. 6(e)], STANet consistently exhibits superior reconstruction performance across various spectral bands.

Fig. 7 presents comparative results from the YRE dataset. It can be observed that STANet demonstrates strong performance in the MSE maps, UQI maps, and spectral curve plots. In the local zoomed-in images [Fig. 7(c)], it can be observed that STANet reconstructs better texture details. STANet demonstrates superior performance over the comparison methods across four evaluation strategies on three datasets, highlighting its strong robustness.

3) Natural Images Experiment Results: The comprehensive experimental results on CAVE and Ntire2022 datasets prove that our proposed STANet model significantly outperforms the comparative state-of-the-art methods in both quantitative and qualitative aspects. On the CAVE dataset, STANet surpasses the suboptimal method, MST++, with a margin of 0.8954 dB and 0.0026 in terms of the PSNR and SSIM metrics, respectively. Similar phenomena were witnessed on the Ntire2022 dataset. Detailed results are visible in the supplementary material.

C. Remote Sensing Land-Cover Classification

In order to further investigate the spectral information of the reconstructed HSI and their ability in image classification applications, additional experiments were conducted using the

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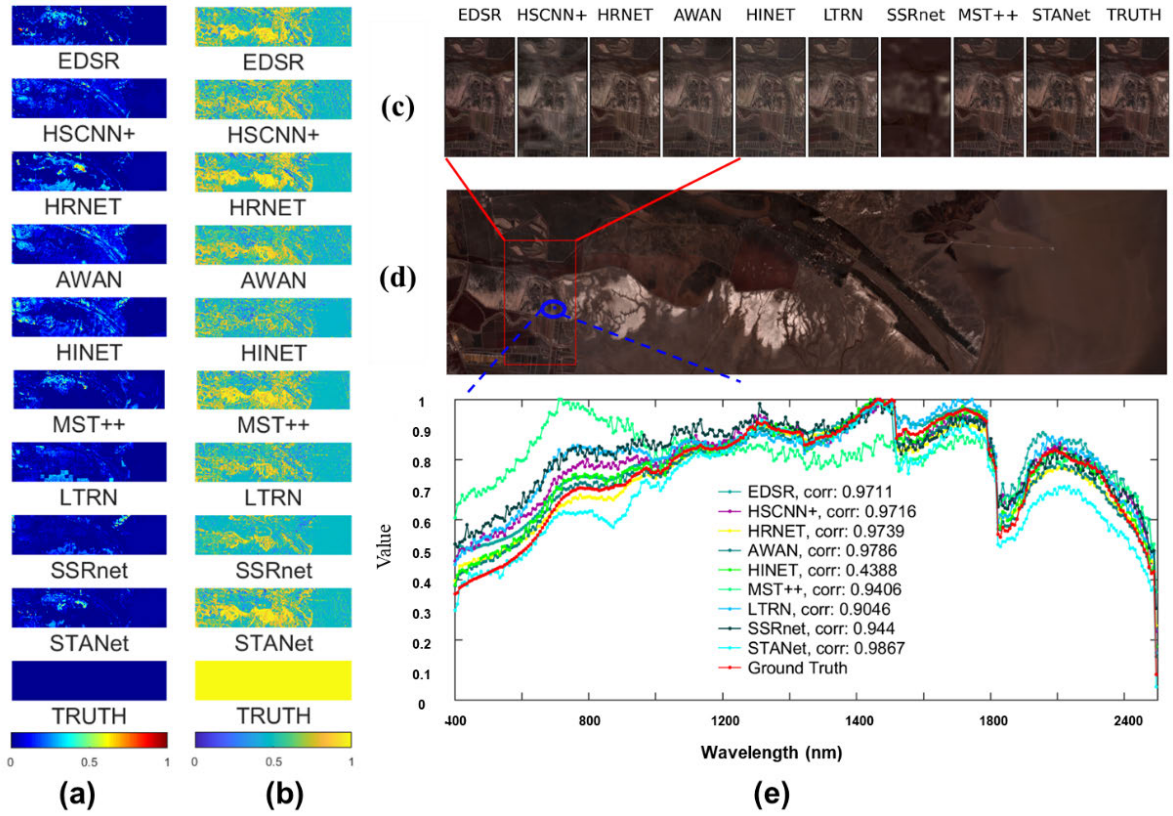


Fig. 7. Qualitative results of the YRE dataset. (a) MSE maps and (b) UQI maps for the spectral band (the 160th band is used here for example). The local zoom-in images (c) created by combining bands of 12, 22, and 37 as the R/G/B channels from the constructed HSI. (d) Displays the original RGB image. The spectral curve (e) of the blue region in recovered and original HSI, as well as the correlation coefficient with the reference spectral curve.

Chikusei dataset. The classifier is support vector machine (SVM). The 20% of the samples were used for training, while the remaining 80% of the samples were used for validation. Ten testing image blocks comprising 16 categories were selected, and specific details are shown in Table VI. The data used for land-cover classification include images obtained from STANet and the other seven aforementioned comparative methods; the original HSI and RGB images are from Chikusei dataset.

In Fig. 8, STANet's reconstructed images demonstrate significant advantages in classification results, showing fewer misclassification and noise compared to other methods, such as the rice field (grown) in scenes Fig. 8(a) and (b), the rice field (first stage) in scene Fig. 8(c), and the forest in scene Fig. 8(d). In some scenes, STANet's classification results even surpass the original hyperspectral data, such as in scene Fig. 8(b) where it correctly identifies bare soil around farmland as "bare soil (farmland)" instead of "bare soil (park)."

The classification accuracy of both reconstructed and original HSI using SVM, including overall accuracy (OA), the Kappa coefficient, and F1 scores, is provided in Table VII. Among the reconstructed HSI, STANet exhibits the optimal classification accuracy, achieving an OA of 91.14 and a kappa coefficient of 89.68. In the F1 scores results, STANet's classification performance is the best among all compared methods, with exceptional performance observed for four specific land-cover categories: "Bare soil (school)," "weeds in farmland," "forest," and "Manmade (red)." It is worth noting that the classification accuracy of the original HSI is inferior to that of STANet. This could be attributed to the following reasons. First, the utilization of SC and spectral constraint

TABLE VI
BASIC INFORMATION OF REMOTE SENSING LAND-COVER CLASSIFICATION DATA

Number of samples	Category	Training samples (20%)	Testing samples (80%)
1686	Water	337	1349
992	Baresoil (school)	198	794
286	Baresoil (park)	57	229
2151	Baresoil (farmland)	430	1721
1108	Weeds in farmland	222	886
7248	Forest	1449	5799
5914	Grass	1183	4731
4571	Rice field (grown)	914	3657
1268	Rice field (first stage)	254	1014
2191	Row crops	438	1753
214	Plastic house	43	171
243	Manmade (non-dark)	49	194
56	Manmade (red)	11	45
1040	Manmade grass	208	832
457	Asphalt	91	366
145	Paved ground	29	116

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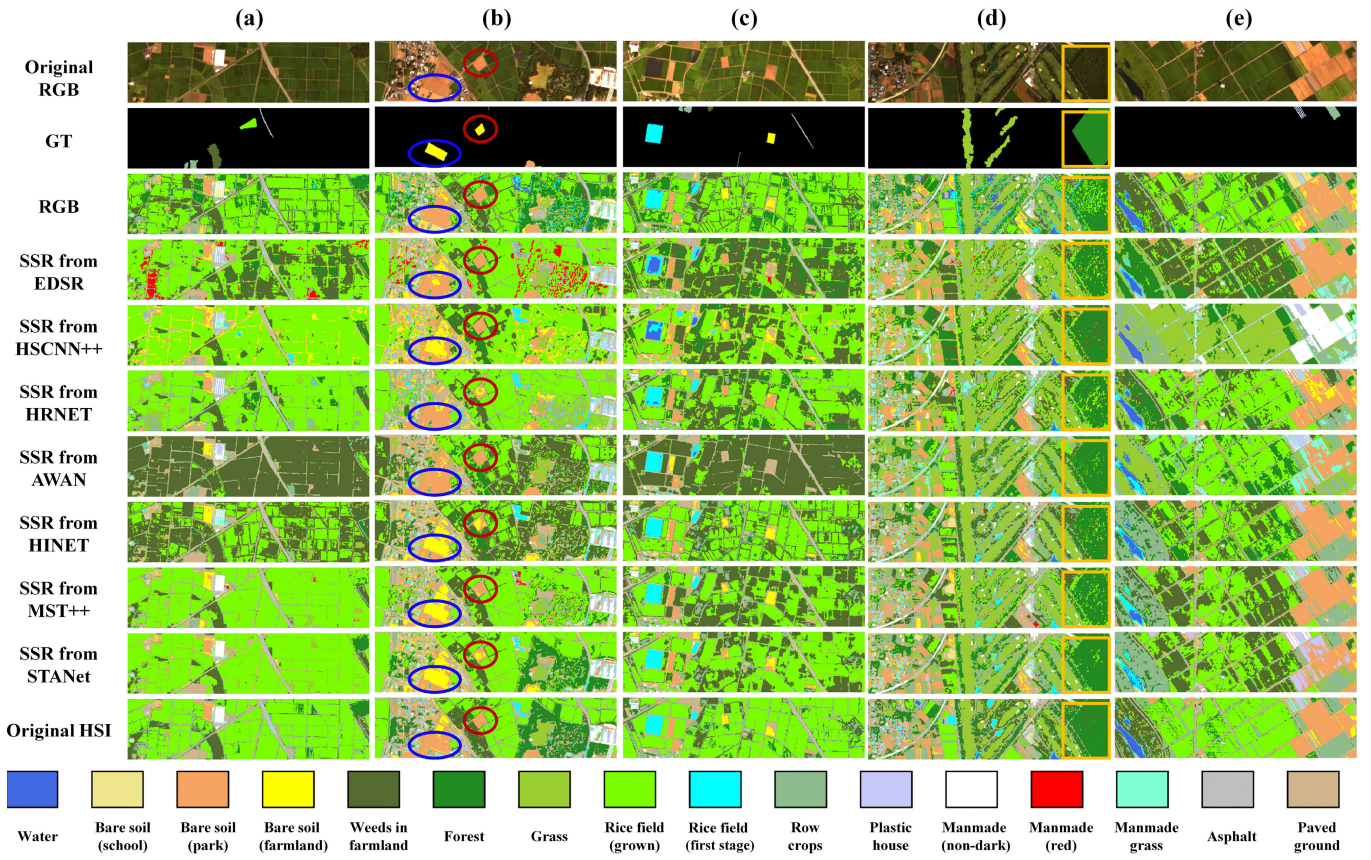


Fig. 8. Land-cover classification results. The first row shows the original RGB images, and the second row displays the classification label files, followed by the classification result images for the reconstructed data from STANet and all the comparative models in the next eight rows. The final row shows the classification result image for the original hyperspectral data. (a)–(e) Patches for different scenarios, respectively.

in STANet enhances the spatial and spectral correlations in the reconstructed HSI. This leads to higher correlation among the same types in the reconstructed images of STANet compared to the original hyperspectral data, as depicted in Fig. 9. In addition to that, by comparing the F1 scores and the land-cover classification results, we found that the main reason for the lower classification accuracy of the original HSI compared to STANet is the poor performance of SVM in classifying bare soil (farmland) (displaying the area circled in Fig. 8), with an F1 score of 41.91, while that of STANet is 89.64.

D. Discussion

In the ablation experiments, we conduct experiments to analyze the impact of the deep fusion block, texture extraction module, spatially guided spectral attention module, and TAM. The ablation experiments are conducted using the PaviaU dataset, and the experimental setup was consistent with the comparative experiments.

1) *Deep Fusion Block:* The deep fusion block is designed to effectively integrate multiple features and facilitate deep feature interactions. We introduced the deep feature fusion structure separately in the texture extraction module and the TAM and conducted comparative experiments by replacing the deep feature fusion structure with a "concatenation" and convolution operation. Table VIII presents the experimental results, demonstrating that the deep fusion block plays a crucial role in the model, particularly in the texture extraction module. When the module is removed, PSNR and SSIM decrease by 0.7669 dB and 0.0038, respectively.
2) *Spatially Guided Spectral Attention Module:* This module employs an improved spectral cross-attention structure to achieve deep aggregation of spectral features. The spectral attention module consists of four parts: spectral constraints, SCs, dense residual block, and the baseline block. We conducted ablation experiments by individually removing each of them. Table IX presents the results of ablation experiments, and it is evident that the spectral attention module has a significant impact on the overall model performance. It is worth noting that the SC had the greatest impact difference among them. PSNR and SSIM decreased by 1.4786 dB and 0.0083 when the SC is removed. This further proves the important role of spatial guidance structure in spectral reconstruction.
3) *Texture Extraction Module:* This module utilizes a learnable texture extraction structure to extract spatial and texture features from the original RGB image. In this module, we mainly use Gabor and Scharr filters for feature extraction. We conducted separate

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TABLE VII
OVERALL CLASSIFICATION PERFORMANCE AND F1 OF EACH SCORE CATEGORY OF DIFFERENT METHODS (%)

	RGB	EDSR	HSCNN+	HRNET	AWAN	HINET	MST++	STANet	Original HSI
Water	98.73±0.34	76.07±1.12	78.73±0.22	97.21±0.17	99.01±0.16	99.85±0.03	99.32±0.03	86.26±0.49	**100±0**
Bare soil (school)	96.55±0.28	93.56±0.79	92.84±0.67	90.67±1.44	98.17±1.04	99.47±0.1	89.26±0.75	**99.6±0.22**	98.79±0.26
Bare soil (park)	30.98±0.37	34.89±1.21	**37±1.27**	20.77±2.15	30.34±0.54	28.67±2.5	29.44±1.56	30.87±1.88	31.32±0.32
Bare soil (farmland)	55.5±2.33	30.82±3.26	70.97±2.96	15.87±8.72	9.59±2.14	**90.07±0.98**	88.78±1.16	89.64±1.58	41.91±7.58
Weeds in farmland	71.65±1.03	47.31±1.11	43.36±1.85	54.58±1.55	47.55±1.05	69.23±1.69	77.65±2.06	**84.62±0.43**	71.78±1.28
Forest	82.54±0.3	88.99±0.27	88.82±0.7	85.57±0.56	96.34±0.28	90.05±0.41	94.27±0.33	**97.55±0.19**	96.88±0.4
Grass	90.36±0.34	89.42±0.13	91.37±0.32	92.04±0.44	91.61±0.42	98.04±0.71	**99.18±0.16**	97.4±0.19	96.35±0.25
Rice field (grown)	77.05±0.77	80.7±0.39	95.52±0.13	81.48±0.28	83.88±0.28	89.49±0.45	90.18±0.29	94.9±0.27	**99.58±0.06**
Rice field (first stage)	96.01±0.46	49.65±2.06	49.82±1.25	90.69±0.41	94.4±0.53	94.17±0.23	93.69±0.24	82.01±0.43	**99.91±0.04**
Row crops	70.7±0.59	49.41±0.65	61.03±2.14	48.14±2.76	75.28±4.2	79.03±3.18	68.22±1.34	78.17±0.5	**80.22±0.4**
Plastic house	55.94±1.76	83.14±3.27	87.65±2.83	52.27±2	**96.4±2.17**	88.27±2.75	62.01±1.58	70.96±3.16	95.28±0.65
Manmade (non-dark)	90.72±0.7	93.17±0.87	83.68±3.5	84.38±3.56	91.94±3.59	93.03±2.92	94.14±1.6	90.01±3.14	**94.15±1.86**
Manmade (red)	**100±0**	90.6±9.98	98.13±3	92.16±4.19	98.89±0.79	99.09±1.5	74.83±9.11	**100±0**	**100±0**
Manmade grass	88.79±1.65	84.9±0.56	80.12±0.41	75.43±1.45	98.18±0.31	96.79±0.94	88.89±0.83	93.48±1.62	**98.33±0.56**
Asphalt	57.75±1.85	67.21±1.28	55.38±1.39	49.19±2.72	83.98±3.39	81.33±1.06	77.9±3.87	81.65±0.47	**98.17±0.33**
Paved ground	**60.95±5.22**	20.1±0.53	24.75±1.54	22.81±1.07	17±2.32	21±2.87	24.05±0.8	25.76±0.78	22.17±2.04
OA↑	80.22±0.42	75.13±0.39	79.95±0.32	75.56±0.5	82.27±0.91	89.43±0.44	89.5±0.23	**91.14±0.21**	89.25±0.52
Kappa↑	77.2±0.47	71.32±0.43	76.95±0.36	71.98±0.55	79.59±1.03	87.76±0.5	87.79±0.27	**89.68±0.24**	87.57±0.59
F1↑	76.48±0.44	67.46±0.41	71.16±0.82	65.77±0.53	75.74±0.58	82.29±0.71	78.19±0.48	81.39±0.40	**82.76±0.31**

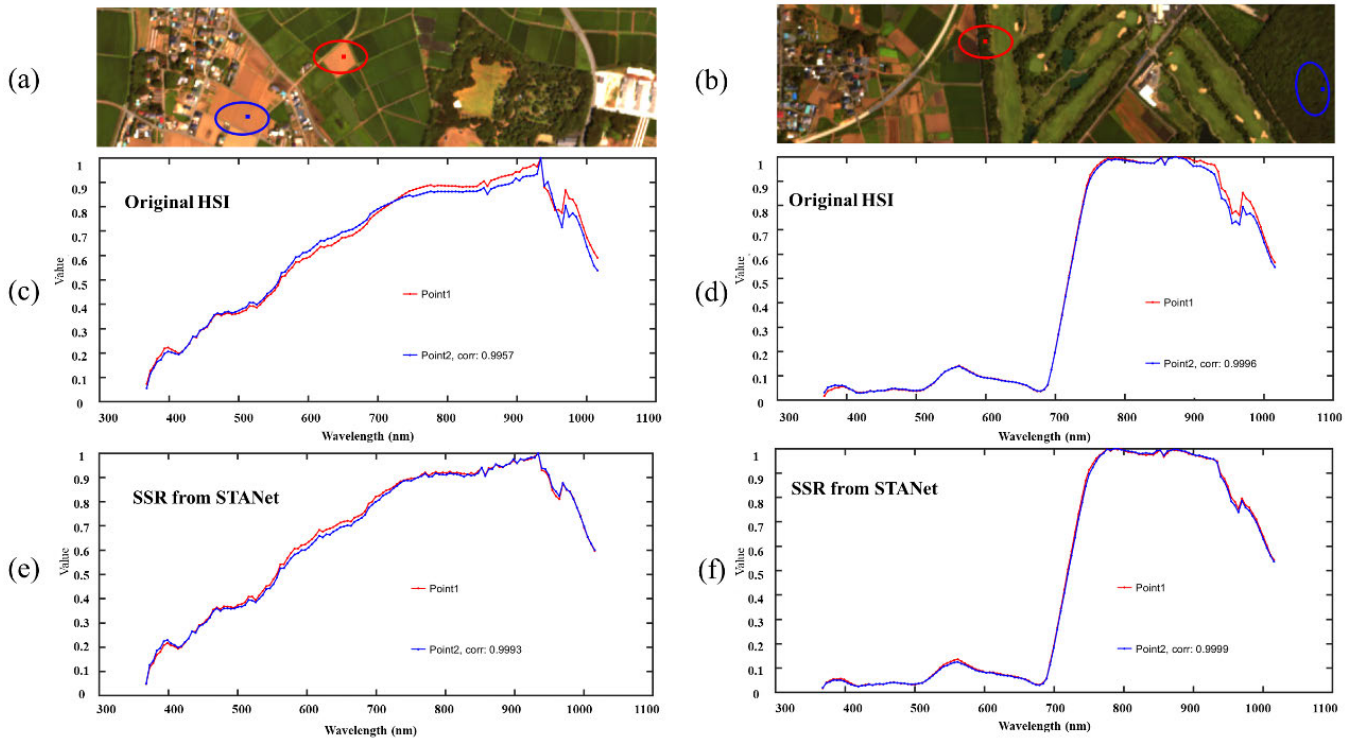


Fig. 9. (a) and (b) Points of interest and the corresponding spectral curve plots of (c) and (e) Bare soil (farmland) and (d) and (f) forest.

TABLE VIII
DEEP FUSION MODULE ABLATION STUDIES

In texture attention module	In texture extraction module	PSNR↑	SSIM↑	ERGAS↓	SAM↓	MSE↓
×	√	32.8169	0.9774	1.7099	4.2343	0.00055
√	×	32.7678	0.9772	1.715	4.2417	0.00055
√	√	**33.5347**	**0.981**	**1.5633**	**3.885**	**0.00046**

ablation experiments for these two filters and replaced them with convolution operations and skip connections. Table X shows the ablation experiments for the Gabor and Scharr filter and the trainability of the Gabor. It can be observed that when Gabor and Scharr filters are removed individually, PSNR decreased

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TABLE IX
SPECTRAL ATTENTION MODULE ABLATION STUDIES

Baseline block	Spatial constraints	Spectral constraints	Dense residual block	PSNR↑	SSIM↑	ERGAS↓	SAM↓	MSE↓
×	×	×	×	32.0511	0.9729	1.8616	4.4169	0.00065
√	×	√	√	32.0561	0.9727	1.8712	4.4304	0.00065
√	√	×	√	32.5321	0.9758	1.7637	4.2508	0.00058
√	√	√	×	32.6152	0.9764	1.7433	4.2569	0.00057
√	√	√	√	**33.5347**	**0.981**	**1.5633**	**3.885**	**0.00046**

TABLE X
TEXTURE EXTRACTION MODULE AND TAM ABLATION STUDIES

Texture attention module	Texture extraction baseline module	Gabor filter	Scharr filter	Texture extraction trainability	PSNR↑	SSIM↑	ERGAS↓	SAM↓	MSE↓
×	√	√	√	√	32.5828	0.9763	1.7516	4.309	0.00057
√	×	×	×	×	32.7741	0.9773	1.7109	4.2256	0.00055
√	Only 1/4 scale	√	√	√	32.4062	0.9751	1.7868	4.7072	0.00060
√	Only 1/2 scale	√	√	√	32.8511	0.9777	1.6952	4.3614	0.00054
√	Only 1 scale	√	√	√	33.3076	1.6029	1.6029	3.9739	0.00048
√	√	√	×	√	32.3618	0.9746	1.804	4.3034	0.00061
√	√	×	√	√	32.5836	0.976	1.7499	4.2166	0.00057
√	√	√	√	×	32.201	0.9737	1.8405	4.3805	0.00063
√	√	√	√	√	**33.5347**	**0.981**	**1.5633**	**3.885**	**0.00046**

TABLE XI
COMPARISON OF COMPUTATIONAL EFFICIENCY RESULTS

Model	MACs (G)	Params (M)	Time (s)
EDSR	20.13	2.46	51
HSCNN+	12.27	6.81	92
AWAN	33.58	4.11	52
HRNET	20.82	31.75	51
HINET	46.26	57.61	57
MST++	12.27	6.81	92
STANet	15.72	8.81	81

by 0.9511 and 1.1729 dB, respectively. The experiments demonstrated that accurately extracting texture information significantly improved the reconstruction performance. Additionally, the results exhibit that as the scale of the feature maps increases, there is an improvement in the model's accuracy, indicating that larger-scale feature maps contribute more to the performance of the model.

4) *TAM:* This module uses the deep fusion block and attention structure to adaptively extract texture and spectral information. We conducted experiments to remove this module to validate its effectiveness. The results of the ablation experiments in Table X demonstrate that PSNR and SSIM decreased by 0.9519 dB and 0.0047 without TAM. This highlights its effective role in fusing texture and spectral features for image reconstruction.
5) *Computational Efficiency:* Multiply-accumulate operations (MACs), number of parameters (Params), and time cost for the STANet and comparative methods are given in Table XI. The results indicate that STANet falls into the moderate range for these three metrics compared to the other methods, with MACs of 15.72, parameters (Params) of 8.81 million, and time cost of 81 s.

IV. CONCLUSION

This article proposed an efficient SSR network, STANet, which can generate HSI from RGB images. Specifically, we first propose a multiscale learnable texture feature extraction module, designed to effectively capture the texture and spatial features in the original RGB images. Second, a spatially guided cross-attention module leverages spatial-spectral constraints is introduced to enhance the efficacy of spectral feature reconstruction. STANet facilitates more effective feature fusion by enabling deep interactions among features. The robustness and superiority of STANet over the state-of-the-art have been illustrated through SSR experiment results obtained from three different datasets. Additionally, we conduct land-cover classification experiments to further corroborate the dependability of STANet's reconstruction outcomes.

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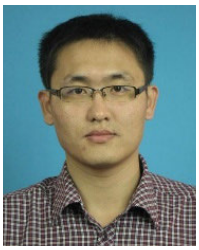
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